

Abstract No.

## Defining Epigenetic Vulnerabilities and Response Programs in IDH-Mutant Gliomas

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IDH-mutant gliomas are epigenetically dysregulated tumors with limited therapeutic options at recurrence. Identifying chromatin-regulatory dependencies specific to the IDH-mutant context may reveal new treatment strategies.

Using a chromatin-focused CRISPR/Cas9 knockout library screen, we identified RBBP7 as a potential IDH-mutant-specific dependency. Functional validation and RNA sequencing were performed in isogenic IDH-wildtype and IDH-mutant glioma cells and extended to 3D patient-derived IDH-mutant glioma stem-like models, revealing context-dependent effects of RBBP7 loss on proliferation, differentiation, and inflammatory programs.

In parallel, an epigenetic drug screen identified KDM6A/6B as candidate therapeutic vulnerabilities. Following pharmacological confirmation, CRISPR/Cas9-mediated single- and double-knockout of KDM6A and KDM6B was performed in 3D patient-derived IDH-mutant glioma stem-like models. Functional characterization and RNA sequencing defined the transcriptional and phenotypic consequences of KDM6A/6B loss. Comparative analysis of RBBP7 and KDM6A/6B perturbation signatures is ongoing to determine whether these vulnerabilities converge on shared response pathways.

To extend translational relevance, patient-derived glioma organoid models are being established, and perturbation signatures will be integrated with public glioma datasets, including TCGA, CGGA, and GLASS, to assess clinical correlates such as recurrence and progression.

Together, this work supports the identification of epigenetic dependencies in IDH-mutant glioma and provides a framework to understand how chromatin perturbations reshape tumor cell states, with implications for future therapeutic targeting.



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